An error budget for near-infrared fruit calibration: a reference-replicate design rule and a falsifiable attenuation diagnostic

Panlin Li[a], Yuchang Li[a], Yutong Feng[b,c], Longjie Li[b,c], Ya Liu[d] and Hua Huang[a,e,*]

[a] *College of Mathematics and Physics, Xinjiang Agricultural University, Urumqi, 830052, China*
[b] *College of Mechanical and Electrical Engineering, Xinjiang Agricultural University, Urumqi, 830052, China*
[c] *Xinjiang Key Laboratory of Intelligent Agricultural Equipment, Urumqi, 830052, China*
[d] *Urumqi Comprehensive Experimental Station, Xinjiang Academy of Agricultural Sciences, Urumqi, 830013, China*
[e] *Research Center for Numerical Simulation and Data Analysis, Xinjiang Agricultural University, Urumqi, 830052, China*

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ABSTRACT

Near-infrared calibrations for fruit soluble solids content (SSC) or dry matter (DM) are judged by the root-mean-square error of prediction (RMSEP), which holds two variance components that are not the model's. We decompose it into an error budget and deliver two tools: a design rule for the reference replicates a target R^2 demands, and a parameter-free—hence falsifiable—attenuation diagnostic. Both presuppose a leakage-free validation protocol, which we specify and audit. In 633 apples with five destructive determinations each, within-fruit variance is 52.30% of the total (intraclass correlation coefficient ICC = 0.477), capping the R^2 of any predictor confined to whole-fruit information at 0.477 for one face and 0.820 for the five-face mean (point estimates)—a population bound assuming face-level deviations exchangeable and mean-independent of whole-fruit information, an assumption we impose rather than test; six parameter-free point predictions track measurement. In 4,318 kiwifruit rescanned by 2–5 devices, device and fruit×device terms take 17.07% and 30.33% of spectral variance. Across ten pre-specified operating points ($R^2 = 0.097$–0.820) the measured instrument-related component spans 20.6% of its mean, the *fitted* trend only 1.1%, while its share of mean squared error rises 5.7-fold, to 24.7%. Random splitting inflates R^2 by 1.41–1.46×. A pre-registered semi-synthetic test reproduced the predicted decay across 13 replicate levels without crossing the ceiling, so the design rule stands as a lower bound with a 1% empirical margin, verified in one of the two blinded targets it changes. Reported performance must be read against a reference-limited ceiling, not unity.

1. Introduction

Non-destructive prediction of internal fruit quality from near-infrared (NIR) or visible–NIR spectroscopy has accumulated three decades of commercial and academic practice [1]. The criterion of success in this literature is remarkably uniform: a destructive chemical determination serves as the reference value, and RMSEP and R^2 on a prediction set are reported. Typical published figures for apple SSC lie at $R^2 > 0.85$ with RMSEP of order 0.5–1.0 °Brix [1, 2, 3], and comparable levels are reported for sugar content in other crops [4, 5]. The spread of portable and imaging instruments [6, 7, 8] and the arrival of deep-learning models [9, 10]—a line that extends to neighbouring vibrational-spectroscopy fields [11]—have widened the reach of this paradigm, but not changed how it is evaluated.

The metric carries a premise that is almost never examined: **that the reference value itself is error-free**. A destructive determination is, however, also a measurement—it has a sampling location, a preparation protocol and an instrument reading, and therefore a variance of its own. In clinical chemistry and metrology it has long been standard that the uncertainty of a reference method propagates into any method-comparison result, and mature frameworks exist for handling it [12, 13, 14, 15]; errors-in-variables regression and attenuation correction likewise have a systematic literature [16, 17]. In fruit spectroscopy calibration, by contrast, the uncertainty of the reference value is rarely estimated explicitly, and even more rarely used to explain the ceiling on reported performance.

Analytical chemistry has recently begun to treat *the structure of measurement error itself* as an object of study. Ezenarro et al. quantified the covariance and correlation structure of spectral measurement errors across several units of the same miniaturised NIR instrument, and derived from it a correlation index intended to guide preprocessing choices [18]. The same line has since been applied to classification with handheld instruments [19] and extended to a multivariate assessment of the "error quality" of replicate scans [20]. This body of work establishes a distinct contribution type: *no new estimator is proposed*; instead the error structure of an existing measurement practice is characterised and a criterion that others can run on their own data is delivered. **All of it, however, analyses the spectral side**—replicate scans and

*Corresponding author
huanghua@xjau.edu.cn (H. Huang)
ORCID(s): 0000-0003-0615-9402 (H. Huang)

between-instrument differences; **there is as yet no counterpart on the reference-value side**. That is the half this paper supplies: when the reference value comes from destructive sampling, its own within-individual variability likewise has an estimable variance structure, and what that structure defines is a ceiling no model can exceed.

A second, parallel neglected component is **between-instrument variability**. Calibration-transfer research has established beyond doubt that models fail across instruments and has produced a large body of correction methods [21, 22, 23, 3, 24]; but that work concerns *how to repair* transfer, not *how much of the RMSEP reported on a single instrument was always the part that changes when the instrument changes*.

A third issue is not a component of the budget but a precondition for computing it at all. **An error budget partitions a reported RMSEP; if that RMSEP was itself inflated by a leaking validation split, the budget is a partition of the wrong total.** The two questions are therefore not separable: one cannot ask how much of the error belongs to the reference method without first fixing what the error is. That fix is not automatic. Machine learning has by now documented systematically how data leakage inflates reported performance [25, 26]; chemometrics has issued its own warnings [27]; the overfitting caused by grouping (or not grouping) by sampling site has been quantified concretely in soil NIR calibration [28]; and the optimistic bias of cross-validation is especially pronounced at limited sample sizes [29]. What none of that literature settles is *which* grouping level a given dataset requires—a question we find has to be answered layer by layer, and audited rather than assumed.

The position of this paper is therefore not to propose a better predictive model but to answer a question that belongs to analytical chemistry itself: **when the reference analytical method carries its own uncertainty, how good can a calibration built on it possibly be?** Concretely, we take RMSEP apart and ask which parts shrink as the model improves and which parts are essentially unchanged between the model operating points compared here. Two structurally complementary datasets are used to estimate the two components separately—one supplies replicate reference determinations, the other supplies replicate scans of the same fruit across instruments.

What results is an *a priori experimental design tool* rather than an after-the-fact critique. Its core is the design rule of §3.6: once the variance ratio σ_f^2/σ_a^2 has been estimated on a small subset of the target batch, one can invert it to obtain the minimum number of independent reference determinations per sample required for a target R^2 to be physically attainable. The attainable ceiling itself (§3.1) is thus a worked example of that rule rather than the end point. The rule is accompanied by a second tool, an attenuation-ratio diagnostic that any laboratory can run on its own data. Both presuppose a validation protocol that does not leak, which is why we also specify one and audit it layer by layer.

Methodologically, this paper proposes neither a new regressor nor a new preprocessing method; it applies variance-component tools already mature in chemometrics to *the reporting of calibration performance itself*. The key design choice is to make the variance-component model deliver **point predictions that contain no free parameters**, so that the decomposition is falsifiable rather than merely descriptive—this is also what separates the present work from the existing "report the measurement uncertainty" genre.

2. Materials and methods

2.1. Datasets

Two datasets carry the error budget; their key properties are summarised in Table 1. They stand in a relation of *division of labour*, not mutual corroboration: the per-face replicate structure unique to the apple data is used to estimate the reference-side component, and the cross-device replicate scans unique to the kiwifruit data are used to estimate the measurement-side component. Public datasets are typically "one fruit, one spectrum, one label" and structurally cannot yield the former.

A **third** dataset appears once and is declared here so that the count is not misread: the public time-series hyperspectral set of post-harvest ripening sequences (five species, 3,405 VIS spectra) used in §3.5 and the Supplementary Material. It contributes *nothing* to either variance component and to no claim the paper makes; it is the dataset on which we initially read a day-level leakage effect and then withdrew that reading. It is kept because the trap it illustrates is worth reporting, not because it supports a result.

Kiwifruit data (measurement-side component). A public dataset [30] of 5,418 fruit and 11,982 spectra over 402–1065 nm (the common usable range of the five devices), labelled with SSC and DM, released on figshare under CC BY 4.0 and first described by Wohlers et al. [31]. We use it as deposited; no augmentation from that work is applied here. **Not every fruit is scanned by several devices**: 4,318 fruit are rescanned by 2–5 devices (only 90 by all five), while 1,100 fruit are scanned by a single device and contribute nothing to the instrument component; the instrument-side analysis uses only the former.

2.2. Variance-component model

For the per-face reference values we use a one-way random-effects model

$$\begin{aligned} &Y_{ij} = \mu + a_i + e_{ij}, \\ &a_i \sim (0, \sigma_a^2), \quad e_{ij} \sim (0, \sigma_f^2), \quad j = 1, \ldots, 5, \end{aligned} \tag{1}$$

Table 1
Data card. The two datasets carry different component-estimation tasks in this paper.

	Apple (reference-side component)	**Kiwifruit (measurement-side component)**
Source	Collected by Xinjiang Agricultural University in 2025; internal and previously unpublished—*not* released with this paper, available from the corresponding author on reasonable request	Public dataset, deposited on figshare under CC BY 4.0 [30] and first described by Wohlers et al. [31]
Origin / batch	Three origins (Gansu, Shandong, Xinjiang); one 2025 collection†	Origin not distinguished
Sample size	633 fruit $\times$ 5 faces = 3,165 reference determinations after cleaning; 3,149 rows remain after fruit–face pairing with the spectra (16 faces have no matching spectrum). **The variance-component analysis uses the former (fully balanced); spectral modelling uses the latter**	5,418 fruit / 11,982 spectra
Key replicate structure	**Five sampling faces per fruit**: top, bottom and three equatorial side faces spaced 120° apart	**Cross-device replicate scans**: 4,318 fruit scanned by 2–5 devices (90 by all five); a further 1,100 fruit by one device only
Reference determination	After destructive sampling, a digital refractometer gives **one independent reading per face** (nominal accuracy ± 0.2 °Brix)	SSC and dry matter (DM)
Spectra	Reflectance, 346 bands over 391–1001 nm	222 bands over 402–1065 nm (common usable range of the five devices; the 1068–1137 nm columns are absent on *all* five devices and were removed before this range was formed)
Cleaning rule (fixed before analysis)	Remove *whole fruit* containing any face with SSC > 20 °Brix (the physiological ceiling is about 18–20, so higher values are entry errors); **do not remove** values < 5 °Brix—locally unripe or rotten tissue genuinely reaches 3–5, and removing them would shrink the finding a priori	Initial preparation drops rows with missing values and removes the wholly missing 1068–1137 nm columns, leaving 222 bands; a later missing-channel mask removes 0 of those 222. The instrument-side analysis then uses only the subset scanned by ≥ 2 devices; fruit scanned once contribute nothing to the instrument component
Use in this paper	Within-fruit reference variability, attainable ceiling, $1/m$ design rule, attenuation diagnostic	Between-instrument error budget, comparison of two model operating points

†The acquisition record states the collection year and the three origins; it does not state whether the material forms a single acquisition batch, so we do not claim it.

where a_i is the whole-fruit deviation of fruit i and e_{ij} the deviation of its j-th face from that fruit's mean. The design is balanced with exactly five faces per fruit, so the classical ANOVA moment estimators $\hat{\sigma}_f^2 = \mathrm{MS}_{\mathrm{within}}$ and $\hat{\sigma}_a^2 = (\mathrm{MS}_{\mathrm{between}} - \mathrm{MS}_{\mathrm{within}})/5$ are used. REML is not used: moment estimators introduce no distributional assumption and no shrinkage, giving the most conservative and least assailable version [32]. The ICC and the minimal detectable difference derived from it are established tools in clinical and biomechanical measurement [33, 34]; agreement analysis in method comparison (Bland–Altman and its reporting conventions) provides a usable framework [35, 36], and the intrinsic limitations of judging agreement by a correlation coefficient have been pointed out repeatedly [37]. All confidence intervals are obtained by bootstrap resampling of **fruit** (the five faces of a fruit move together, $B = 2000$).

If the reference value is taken as the mean of m faces, $\bar{Y}_i^{(m)}$, then any model that can perceive only whole-fruit information (i.e. can predict only $\mu + a_i$) satisfies

$$R^2 \leq \frac{\sigma_a^2}{\sigma_a^2 + \sigma_f^2/m}, \qquad \mathrm{RMSE} \geq \frac{\sigma_f}{\sqrt{m}}. \tag{2}$$

Equation (2) is a statement about the **population** R^2, i.e. the variance ratio itself. The R^2 we *report* is a finite-sample score whose total sum of squares is referenced to the training-fold mean (§2.3); by the identity given there that denominator is never smaller than the test-mean one, so the reported score is never lower and is **not** guaranteed to obey Eq. (2) exactly in finite samples. On our splits the two denominators differ by a factor of 0.996–0.999, i.e. at most 0.001 in R^2, so every comparison we draw between a measured score and this ceiling is approximate to that order and we do not claim the bound as an exact property of the reported statistic. **The same reading applies to all six point predictions of §3.2 and to the design rule of §3.6**: each is a closed-form population quantity compared against a training-centred empirical score, so agreement is to be read at the same 0.001

order, not as an exact identity. Equation (2) requires **two** conditions, and it is worth separating them. The first is exchangeability of the five faces—the face-level deviations e_{ij} are mutually uncorrelated within a fruit, so that the variance of their mean is exactly σ_f^2/m. The second is that the face deviations carry **no information the predictor can see**. Stating this second condition as zero covariance, $\mathrm{Cov}(a_i, \bar{e}_{i\cdot}) = 0$, is **not sufficient** for the bound as written. Zero covariance constrains only linear dependence, whereas the cross-term that must vanish when a predictor g is subtracted is $\mathbb{E}[(a_i - g)\,\bar{e}_{i\cdot}]$, and for nonlinear g that expectation is not controlled by a covariance. A construction makes the gap concrete: let a scalar whole-fruit signal be $S \sim U[0, 2\pi]$ and let the five face deviations be a random permutation of $\cos S, \ldots, \cos 5S$. They are mutually uncorrelated and each uncorrelated with S, yet their mean is a deterministic function of S and is therefore predictable from whole-fruit information alone. **The condition we require, and state, is the stronger one**: the mean face deviation is *mean-independent* of every whole-fruit quantity a predictor can condition on, $\mathbb{E}[\bar{e}_{i\cdot} \mid a_i, S_i] = 0$, where S_i denotes the spectrum. Under zero covariance alone Eq. (2) still holds, but only against predictors that are affine in the whole-fruit information—not against "any" predictor. Exchangeability alone is likewise not enough: with $a_i = e_{i1} + e_{i2}$ the deviations are mutually uncorrelated yet perfectly correlated with their own mean, and Eq. (2) fails. Both conditions are **assumptions, not consequences** of writing the model down. The first is what we test in §3.2: a permutation test of compound symmetry cannot reject it. The second is not testable on this dataset at all—not the covariance version and still less the mean-independence version—because it would need technical replicates on the same face (§4.3(2)); we impose it. Under merely *approximate* mean independence the expression is an approximation whose error is of the order of the residual predictable part of $\bar{e}_{i\cdot}$.

A precise definition of terms. The m "replicates" of this paper are *destructive samplings at m different positions on the same fruit*, not m replicate analyses of the same sample and the same measurand. Correspondingly σ_f^2 is the **spatial-subsample variability defined by our five-point scheme**, pooling genuine between-face chemical heterogeneity, sampling and preparation variability, and refractometer reading error (they are inseparable here, §4.3(2)). The accurate reading of Eq. (4) is therefore "*how many approximately independent spatial subsamples per fruit are needed to estimate whole-fruit mean quality to a given precision*", not "how many parallel determinations the reference analytical method must perform". The two are mathematically of the same form but correspond to different laboratory operations: the former requires moving the sampling location, the latter only re-measuring the same homogenate.

2.3. Modelling and validation

Unless stated otherwise, partial least squares regression (PLS) is used, with the number of components selected by grouped cross-validation within the training set (the "weak model" operating point of §3.4 is a deliberate exception, with the component count fixed at 10 in order to construct a poorly performing comparison point). The total sum of squares in R^2 is referenced to the **training-set** mean, not to the test set's own mean, because the training mean is the only baseline actually available at prediction time; the test set's own mean is an oracle baseline. We note the direction of this choice explicitly: since $\sum_i (y_i - \bar{y}_{\text{train}})^2 = \sum_i (y_i - \bar{y}_{\text{test}})^2 + n(\bar{y}_{\text{test}} - \bar{y}_{\text{train}})^2$, the training-mean denominator is never smaller, so this convention yields an R^2 that is never *lower* than the test-mean convention. Replaying every split used in this paper without refitting, the ratio of the two denominators has mean 0.996–0.999, so the two conventions differ by at most 0.001 in R^2 — below the precision at which we report it. Except for the row-wise random split used *deliberately as a leakage control* in §3.5, every performance estimate is **grouped by individual**: multiple spectra of the same fruit never straddle the train/test boundary. For the time-series spectra a further layer of **grouping by acquisition day** is added (§3.5). Preprocessing compares raw, SNV, Savitzky–Golay first derivative and their combinations [38, 9]. These practices follow accepted best practice in quantitative spectral modelling [39, 40].

Replication and uncertainty reporting. Everything that depends on a random split is run in **two stages**: feasibility is first verified on a single development seed (Pilot), then the whole analysis is rerun on five seeds fixed in advance, {20060515, 20041210, 19810915, 2023, 2 (Formal); **every split-dependent quantity reported in the text is the Formal result**, and the Pilot values do not enter the text. It should be stated explicitly that a second class of quantities *does not depend on any random split*—the variance components and ICC, the attainable ceiling, the label-only baseline (all closed-form moment estimators), and the spectral-quality and saturation diagnostics (no split at all). These are seed-independent, so the original computation was retained and not rerun; this is noted at each point of use below. Under each seed, 20 repeated fruit-grouped holdouts are performed ($n_{\text{primary}} = 20$) and the component count is selected by 4-fold grouped cross-validation within the training set ($K_{\text{inner}} = 4$), so each R^2 in the main apple table and in the leakage controls aggregates $5 \times 20 = 100$ **repeated** splits (repeated splits share data and are not independent samples). Replication counts for secondary analyses differ and are labelled at each occurrence: 15 repeats per seed for the band-subset analysis (75 in total), and 6 repeats per seed for the strong kiwifruit model, the time-series day-index controls and the semi-synthetic degradation experiment of §3.7 (30 in total each). The

20-repeat figure above therefore describes the main apple table and the leakage controls, not every analysis in the paper. All confidence intervals use a two-level **cluster bootstrap with seeds as clusters** (seeds are resampled first, then repeats within a seed, $B = 2000$): the 20 repeats under one seed share a single random stream and are not independent samples, and an i.i.d. bootstrap systematically underestimates interval width. The reported SDs reflect only the variance introduced by splitting and modelling, not hardware jitter. Every ratio-valued quantity (attenuation ratio, split-inflation factor, RMSE ratio) is computed as a **ratio of means** $\mathbb{E}[X]/\mathbb{E}[Y]$ rather than a mean of ratios $\mathbb{E}[X/Y]$: with a denominator as small as $R^2 \approx 0.06$ the latter is pulled up appreciably by a few very small denominators. We do *not* claim a general direction for this discrepancy—for dependent X and Y neither ordering holds universally—but report it as measured: on our data the mean of ratios reaches 2.28 against 1.41 for the ratio of means (see the companion workbook). The decisive reason for the choice is nonetheless structural rather than empirical: theoretical predictions such as Eq. (3) are themselves ratios of expectations, so only the ratio of means is of the same form. All models here are PLS and closed-form moment estimators with no tensor operations; computation runs on CPU and is reproducible given the seed.

2.4. Pre-registration and traceability

The statistical analysis plan, the cleaning rules and every criterion were written down and archived before the final results were seen. **Every variance component, R^2, RMSE, ratio and confidence interval** reported here is produced by a script and written to a workbook, and each such number in the figures is traceable to a specific table (dataset metadata such as sample sizes and instrument models are traceable to the data card instead). Quantities that the text reports but derives from other stored components—the variance ratio σ_f^2/σ_a^2, and the confidence-interval widths and their ratio in the Supplementary Material—are themselves exported, so that "traceable" means *stored*, not merely *recomputable*. What is written to the workbooks are these *statistics* themselves; process-level intermediates such as the $B = 2000$ bootstrap resamples are not released with the workbooks and can be replayed from the scripts and seeds when required. Any deviation from the analysis plan is entered in the protocol's deviation log.

3. Results

3.1. Within-fruit variability of the reference value and the resulting ceiling

After cleaning, $n = 633$ fruit, $\hat{\sigma}_a^2 = 2.327$ and $\hat{\sigma}_f^2 = 2.552$: the **within-fruit (between-face) variance is 52.30% of the total** (ICC = 0.477, bootstrap 95% CI [0.436, 0.515]). The **within-fruit (between-face)** SD is $\hat{\sigma}_f = 1.597$ °Brix and the standard error of the five-face mean is $\hat{\sigma}_f/\sqrt{5} = 0.714$ °Brix. (The SD of the single-face *values* themselves is larger, 2.207 °Brix, because it also carries the between-fruit component $\hat{\sigma}_a$; the two must not be confused.) The median within-fruit range is 2.80 °Brix, with 44.1% of fruit exceeding 3 °Brix and 20.7% exceeding 5 °Brix (Fig. 1a). For scale, the median within-fruit SD across five faces, 1.110 °Brix, is 5.6 times the refractometer's nominal accuracy.

The proportion differs markedly between the three origins (no formal hypothesis test of origin differences is performed here; point estimates only): 38.5% for Shandong, 52.5% for Xinjiang and 66.5% for Gansu (Fig. 1b), corresponding to attainable R^2 ceilings from 0.615 down to 0.335—nearly a factor of two. The ceiling is therefore not a species constant and **must be reported per batch**.

By Eq. (2), with a single face as the reference value the R^2 ceiling is 0.477; with the mean of five faces it rises to 0.820 (Fig. 1c). **These are point estimates, and we quote them as such throughout the paper, including in the abstract and conclusions.** Propagating the ICC confidence interval [0.436, 0.515] through Eq. (2) gives [0.436, 0.515] for a single face and [0.794, 0.842] for the five-face mean. A reader who wants the conservative reading—the one under which a model is hardest to declare ceiling-bound—should take the upper limits, 0.515 and 0.842, and every statement we make about this ceiling should be read as carrying that interval.

3.2. Six point predictions: theory versus measurement

Equation (2) and its corollaries give four groups totalling six **parameter-free** point predictions that can be checked directly against the data (Fig. 2; all predicted and measured values are in the companion workbook):

1. $1/m$ **scaling**: taking the mean of the **first** m **sampling faces** of each fruit (in the fixed order top, bottom, side 1, ...), the measured variance deviates from the theoretical value $\sigma_a^2 + \sigma_f^2/m$ by -1.61% ($m = 2$), $+2.30\%$ ($m = 3$) and $+0.72\%$ ($m = 4$). A fixed subset is used rather than the average over all $\binom{5}{m}$ combinations, so the sensitivity of this test to subset choice has not been assessed. Note that $\hat{\sigma}_a^2$ and $\hat{\sigma}_f^2$ are estimated once from the complete $m = 5$ design and that **no additional fitting whatsoever** is done at $m = 2, 3, 4$, so this is not circular; it remains, however, an in-sample extrapolation on the same 2025 apple collection rather than validation on an independent sample.

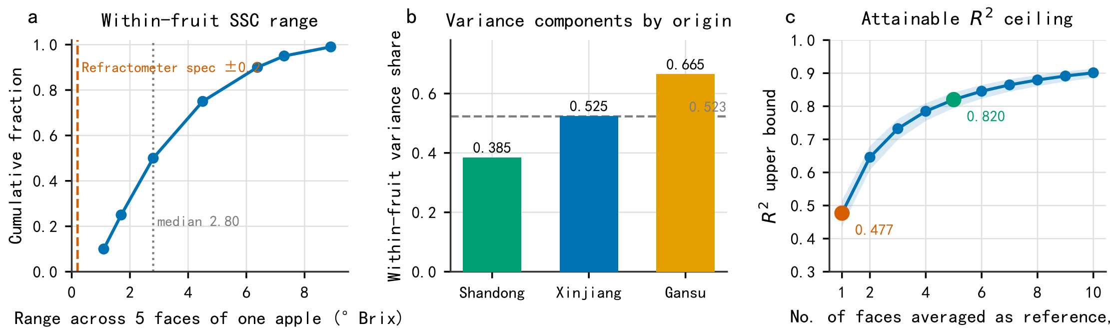

Figure 1: Within-fruit variability of the reference value. (a) Cumulative distribution of the SSC range across the five faces of a fruit; dashed lines mark the refractometer's nominal accuracy and the median range. (b) Within-fruit share of the total variance by origin; the horizontal dashed line is the pooled value 0.523. (c) Attainable R^2 ceiling as a function of the number of faces m averaged into the reference value; the shaded band is the bootstrap 95% interval of the ICC.

2. **Exchangeability across faces**: the mean off-diagonal element of the covariance matrix is 2.331, consistent with the value $\hat{\sigma}_a^2 = 2.327$ predicted under exchangeability; a permutation test of compound symmetry cannot be rejected ($p = 0.261$). The data are therefore **compatible** with the $1/m$ scaling; failure to reject the null is not proof that the scaling holds exactly.
3. **Label-only baseline**: predicting the SSC of one face from the mean of the other four faces of the same fruit has theoretical value $1 - (\sigma_f^2 + \sigma_f^2/4)/(\sigma_a^2 + \sigma_f^2) = 0.3462$ against a measured 0.3456. This expression contains no free parameter and is the sharpest of the group.
4. **Attenuation ratio**: the fourth group is the only one of the four that carries a confidence interval, and is therefore treated separately below.

The attenuation ratio: the one prediction with a confidence interval. Suppose a model perceives only whole-fruit information *and* extracts the same signal under both targets—so that the variance it explains, the numerator of R^2, is the same for both. The ratio of R^2 between the two targets then equals the ratio of their variances. The second half of that supposition is an **assumption in its own right**, not a consequence of the first: two predictors can both be confined to whole-fruit information and still fit different amounts of it. The paired design below—both targets sharing every split—is what makes it plausible, and the interval test is what would expose it if it failed. One subtlety decides what that ratio is. The "whole-fruit" target here is the *mean* of five sampling faces, so it still carries noise $\sigma_f^2/5$; the prediction is consequently *not* the ICC, but

$$\frac{R^2_{\text{face-level}}}{R^2_{\text{whole-fruit mean}}} = \frac{\sigma_a^2 + \sigma_f^2/5}{\sigma_a^2 + \sigma_f^2} = 0.5816. \qquad (3)$$

We test it under a paired design in which the two targets **share the same splits** (5 seeds × 20 fruit-grouped holdouts each, 100 in total). The measured ratio is **0.564**, with a two-level seed-clustered bootstrap 95% confidence interval [0.531, 0.596] that **contains** the prediction 0.5816 of Eq. (3). The comparison is discriminating rather than merely permissive: the ICC itself—0.477, which corresponds to a noise-free whole-fruit target, i.e. the limit $m \to \infty$—falls *outside* that interval. The data thus distinguish between two candidate predictions that differ by only 0.10, and select the one the variance-component model actually implies.

One caveat on the interval's width. Both numerator and denominator are R^2 values of magnitude only 0.06–0.12, so the ±0.03 width is driven mainly by the weakness of the spectral signal itself rather than by unstable splitting.

3.3. Within-fruit variability is not mainly explained by the known calyx–stem gradient

An immediate objection is that the bottom (calyx end) of a fruit is known to be sweeter than the top (stem end), so the "within-fruit variability" above might be nothing but a restatement of that gradient. The gradient is confirmed—the bottom exceeds the top by 0.405 °Brix (paired $t = 4.42$)—but its magnitude is only 36% of the median within-fruit SD, and **the fixed effect of face position explains just 0.95% of the within-fruit variability**. A cleaner control uses the three side faces only: they lie on the same equatorial band, are anatomically equivalent, and contain no stem–calyx gradient at all. The between-face variance estimated from them is 2.4396, i.e. 95.6% of the value 2.5515 estimated from all five faces, and the median range across just those three faces is still 1.80 °Brix. Substantial spatial dynamics of sugars and cell state within apple flesh have direct cytological evidence in **watercored** fruit [41]—a physiological disorder, so we

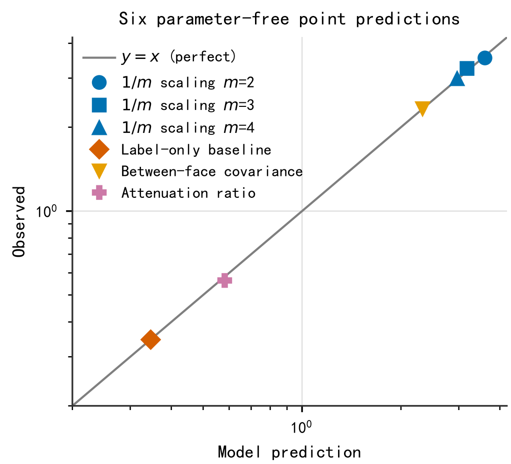

Figure 2: Parameter-free point predictions: abscissa, the value predicted by the variance-component model; ordinate, the measured value; grey line, $y = x$.

do not extend that evidence to sound fruit; this is *compatible* with the dispersion we observe at the reference-value level, but because this dataset has no technical replicates on the same face (§4.3) it cannot be used to attribute that dispersion to biological heterogeneity. To be precise: the face-position effect is not statistically zero (permutation test $p < 0.001$), but its explanatory power is very weak (0.95%). The variability therefore has almost no structure in the spatial variables this dataset *records*—which is not to say it has no biological structure, since it may relate to unrecorded factors such as sun exposure or canopy position; but precisely for that reason, standardising the sampling position alone will not eliminate it.

3.4. Between-instrument variability and its invariance to model quality

In the kiwifruit data there are no technical replicates within a device ($n_{\text{scan}} = n_{\text{device}}$ always), so the "instrument×sample interaction" and "scan/repositioning noise" are **inseparable** and only their sum can be reported. On that basis, the variance of the SNV-corrected spectra decomposes into 52.60% between fruit (biological and batch), **17.07% device main effect** and 30.33% fruit×device pooled residual (Fig. 3a). Nearly half of the spectral variation thus originates in the measurement system rather than in biological difference; of that, the device main effect is a systematic bias which is identifiable under a crossed design and in principle correctable—precisely the target of calibration-transfer methods [21, 23, 3].

The predictive-level result is more direct. We repeat the same analysis at two model operating points: a weak model (SNV, full spectrum, component count fixed at 10) gives $R^2 = 0.393$ and RMSEP $= 1.809$ %DM; a strong model (first derivative, 700–1065 nm, component count selected by cross-validation) gives $R^2 = 0.819$ and RMSEP $= 0.988$ %DM. Their RMSEPs differ by a factor of 1.83, but the part of RMSEP corresponding to the dispersion of predictions for the same fruit across devices barely moves: 0.495 versus 0.490 %DM. Two points, however, cannot distinguish invariance from a shallow trend, so we scanned the operating range. The scan uses a *different fold rule* from the two points just quoted (fruit assigned at random to five folds under each of the five fixed seeds, rather than the deterministic grouped k-fold used above), because a slope needs an uncertainty and the deterministic rule provides no seed dimension. Its numbers are therefore its own and are not interchangeable with the two above. **Ten configurations were fixed in advance** (preprocessing $\in$ {raw, SNV, first derivative, second derivative, SNV + derivative}, spectral range $\in$ {full, $\geq$700, $\geq$800 nm}, component count $\in$ {2, 4, 6, 10, cross-validated}; the two operating points above are configurations W and S of that list) and **all ten are reported**—individually, in Table S1 of the Supplementary Material—each run under the five fixed seeds. They span R^2 from 0.097 to 0.820, over which the total RMSEP falls by 55% (2.207 to 0.986 %DM).

Across that scan the pooled component stays between 0.457 and 0.562 %DM (mean 0.505), while its share of mean squared prediction error rises from 4.3% to 24.7%, a factor of 5.7 (Fig. 3b). Regressing the component on R^2 gives a slope of +0.0073 %DM per unit R^2, with a seed-clustered bootstrap 95% interval of [+0.0038, +0.0109] that **excludes zero**: the component is therefore *not* strictly invariant, and we report that rather than claim invariance. The size of the trend is what matters. The regression accounts for 0.5% of the variance of the component across configurations, and over the entire R^2 range the *fitted trend* predicts a change of +0.005 %DM, i.e. 1.1% of the component's own magnitude, while the denominator it is divided by shrinks by more than half. That 1.1% is the trend, not the observed movement: the component's

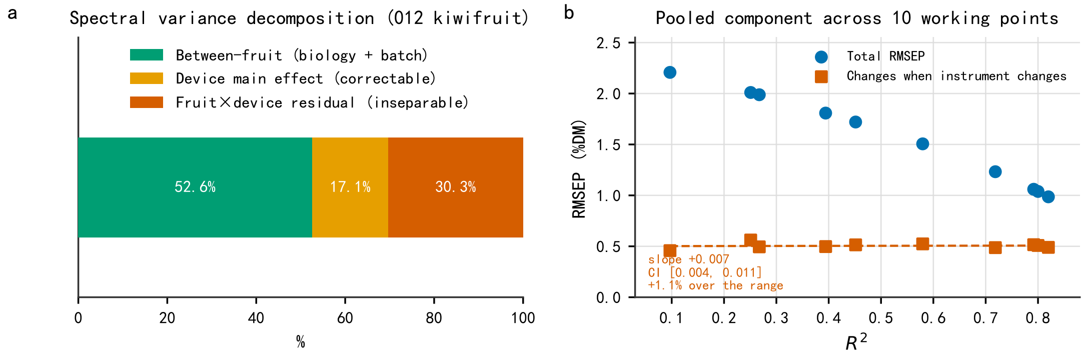

Figure 3: Between-instrument variability on the measurement side. (a) Three-component decomposition of the SNV-corrected spectral variance. (b) Total RMSEP (blue circles) and the *pooled* component that varies across devices (orange squares—device×sample interaction together with scan/repositioning noise, which this design cannot separate; see text) at **ten pre-specified operating points**, against R^2 on the abscissa. The dashed orange line is the linear regression of that component on R^2 and the band is its seed-clustered bootstrap 95% interval.

measured values span 20.6% of their mean across the ten configurations, and the residual SD of ±0.03 %DM shows that almost all of that spread is configuration-to- configuration scatter unrelated to model quality. All five of these fitted diagnostics are exported to the workbook alongside the slope.

Two qualifications still stand. First, this quantity is the **pooled device×sample interaction and scan/repositioning noise**, not pure instrument repeatability (§4.3). Second, the configurations differ *simultaneously* in preprocessing, spectral range and component budget, so "model quality" names that joint change and not any one factor within it; the scan is a sweep of operating points, not a controlled ablation. Within those qualifications the pattern is: model improvement compresses the compressible part of the error and leaves the pooled component very nearly untouched, so the share of the latter rises steeply.

3.5. Validation protocol: grouping is not a single decision

The components above are meaningful only if the validation protocol itself is leakage-free. The necessity of grouped cross-validation and its implementation details have been discussed systematically in the methodological literature [42, 43]; the contribution of this section is to point out that *the grouping level itself must be determined layer by layer from how the data were generated*. This section reports the one leakage we confirmed; a second, day-level reading that we withdrew after a control test is documented in full in the Supplementary Material, because the trap it illustrates is easy to miss.

In the apple data, replacing a row-wise random split by grouping by fruit lowers R^2 from 0.0878 to 0.0625 (face-level SSC) and from 0.1692 to 0.1195 (whole-fruit mean SSC), inflation factors of 1.41 and 1.42; under first-derivative preprocessing the same comparison gives 0.0917 → 0.0631 and 0.1580 → 0.1079, factors of 1.45 and 1.46. Across the four combinations the inflation is 1.41–1.46×, with seed-clustered bootstrap 95% confidence intervals of [1.21, 1.71], [1.27, 1.63], [1.25, 1.67] and [1.30, 1.71] respectively; all four lower limits are clearly above 1 (Fig. 4). The mechanism is that the remaining faces of the same fruit fall into the training set and leak whole-fruit information.

On time-series hyperspectral data (post-harvest ripening sequences of five species, 3,405 VIS spectra) we initially read a second, deeper layer of leakage: predicting the imaging day index of a single spectrum gives $R^2 = 0.887$ when grouped by fruit and 0.705 when grouped by acquisition day. **That reading does not survive a control test and we withdraw it.** The imaging day index is numbered from the acquisition-day label, so holding out a whole day also removes an entire level of the target; rerunning the comparison with destructive measurements *not* derived from the grouping variable—fruit firmness and storage duration—leaves only 2 of 6 combinations above the pre-specified threshold, both on the one target still strongly correlated with acquisition day. The six are two targets crossed with **three evaluation levels—kiwifruit, avocado and the pooled set—not three species**; the remaining three species of that dataset enter only through the pooled level (see the Supplementary Material for why). We therefore **do not claim** a day-level leakage mechanism independent of individual-level leakage. The episode is nonetheless a reproducible validation-protocol trap in its own right: *when the prediction target is derived from the grouping variable, grouping by that variable manufactures a leakage-like performance drop out of nothing*, and the size of the drop alone cannot distinguish the two. The full analysis, the terminal-date decomposition and Fig. S1 are given in the Supplementary Material (§S1).

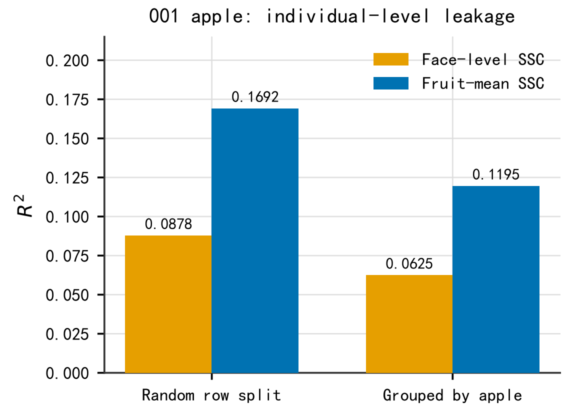

Figure 4: Genuine individual-level leakage in the apple data: a row-wise random split inflates performance relative to grouping by fruit. **The bars plot the raw-preprocessing comparison** for the face-level and the fruit-mean SSC target (factors 1.41 and 1.42); the first-derivative comparison gives 1.45 and 1.46 and is reported in the text of §3.5 rather than drawn here, so the range quoted elsewhere, 1.41–1.46×, spans all four combinations. All values are Formal five-seed results. The withdrawn day-level comparison is Fig. S1 in the Supplementary Material.

3.6. A design table for the number of reference replicates

Inverting Eq. (2) answers the question "how many independent replicate determinations must the reference value have for the attainable R^2 ceiling to reach a target $R^2_{\max}$":

$$m \geq \frac{\sigma_f^2}{\sigma_a^2} \cdot \frac{R^2_{\max}}{1 - R^2_{\max}}. \tag{4}$$

Substituting the apple value $\sigma_f^2/\sigma_a^2 = 1.0965$: $R^2_{\max} = 0.7$ requires $m \geq 3$; 0.8 requires $m \geq 5$; 0.9 requires $m \geq 10$; 0.95 requires $m \geq 21$. Because the ratio already differs appreciably between the three origins in this 2025 collection (§3.1), practical use should first estimate σ_f^2/σ_a^2 on a small subset of the target batch and then fix m accordingly; we did not observe variation across batches and make no claim about how the ratio changes with batch. **Equation (4) gives a structural *lower bound*, not a sufficient condition.** The measurements in §3.7 show that retraining a model on noisy reference values incurs a further loss the formula does not account for: the measured median signed relative deviations are -0.9% and -0.3%, and two of four blinded targets fall short by less than 0.008. A margin above that bound is therefore advisable in practice. The operational rule given in §3.7 is to *discount the clean baseline R^2_{clean} by 1% before inverting*—the 1% here is a *discount factor* and is not the same quantity as the relative deviations just quoted.

3.7. A direct test of the ceiling and the design rule on a dataset with a strong enough spectral signal

Sections 3.1–3.6 all rest on the apple data, whose spectral side performs poorly under the PLS protocols evaluated here (§4.3). One objection therefore has to be met head-on: *if the ceiling of this paper was never approached by any model that actually works, is it merely a mathematical property that is never triggered?* This section answers with a **semi-synthetic experiment** whose criteria were all written down and archived before it was run. In the interest of full disclosure: this is the *second* version of that pre-registration. The first (Appendix G of the protocol) set the ceiling against a variance the model could in fact perceive, and its own hard-falsification criterion failed; it is kept verbatim rather than deleted. The present version corrects that setting—**with no threshold changed**—and its four design targets were fixed before the corresponding replicate levels had been measured.

We switch to the kiwifruit data, whose spectral signal is strong enough (calibration R^2 of 0.827 for DM and 0.861 for SSC), and **inject** reference noise into the labels. For each fruit the original label is replaced by the mean of m virtual replicate determinations—that is, independent noise of variance σ_f^2/m is added. The noise magnitude is set by **splitting** the kiwifruit label variance σ_y^2 into a between- and a within-fruit part in the ratio $\sigma_f^2/\sigma_a^2 = 1.0965$ measured on apples, and injecting the within-fruit part: $\sigma_f^2 = \left[1.0965/(1 + 1.0965)\right] \sigma_y^2 = 0.5230\, \sigma_y^2$, which is exactly the 52.30% within-fruit share of the apple data ($= 1 - \text{ICC}$). We state the split rather than the ratio alone because "scaled to the label variance using the ratio 1.0965" also reads as $\sigma_f^2 = 1.0965\, \sigma_y^2$; the two readings differ by a factor of 2.1 and only the former reproduces the m_{req} values of Table 2. **The model is retrained on the noisy labels** (the component count is also selected using the noisy labels), which is what corresponds to the real situation—the experimenter only ever holds noisy reference values. Thirteen levels of m are used $(1, 2, \ldots, 10, 14, 20, \infty)$ with the Formal five seeds. The

Table 2
Blinded test of the inverted design rule. The shortfall is the measured R^2 minus the target value.

Target	R^2 target	Inverted m_{req}	Measured R^2	Shortfall
DM	0.75	6	0.7536	$+0.0036$
DM	0.78	9	0.7750	-0.0050
SSC	0.75	4	0.7578	$+0.0078$
SSC	0.80	7	0.7989	-0.0011

evaluation regime is the one that produces the values 0.827 and 0.861 in §4.3—fruit-grouped repeated hold-out, 6 repeats per seed, component count selected by 4-fold inner grouped cross-validation. It is **not** the same regime as the 5-fold grouped cross-validation that gives $R^2 = 0.819$ in §3.4, and the two numbers must not be interchanged.

Under this setup the model can perceive the entire clean label (variance σ_y^2) and only the injected σ_f^2/m is imperceptible to it, so the structural ceiling of Eq. (2) becomes $\sigma_y^2/(\sigma_y^2+\sigma_f^2/m)$ and the parameter-free prediction is

$$R^2_{\text{pred}}(m) = \frac{R^2_{\text{clean}}\,\sigma_y^2}{\sigma_y^2 + \sigma_f^2/m}. \tag{5}$$

Results (Fig. 5). At none of the 13 levels of m does the measurement cross the structural ceiling; the median relative deviation between measurement and Eq. (5) is -0.9% (DM) and -0.3% (SSC) (the pre-registered P2 criterion is stated on the absolute value, $|R^2_{\text{obs}}/R^2_{\text{pred}} - 1| \le 15\%$). More notable is the *shape* of the deviation: it is **uniformly negative** and **shrinks monotonically** as m grows (DM from -1.67% to -0.67%, SSC from -1.59% to -0.12%). This agrees with the mechanistic expectation—the less noise is injected, the less of it is fitted away during retraining—and the shape was not obtained by any fitting.

A direct test of the design rule. We specified four blinded targets in advance (the m at which each lies had no measured result at the time of specification), inverted for the required replicate count, and only then measured (Table 2). What is inverted here is Eq. (5), not Eq. (4). Two things differ. In the semi-synthetic setting the model perceives the entire clean label, so the denominator carries σ_y^2 rather than σ_a^2; and the clean baseline itself falls short of 1, which must be subtracted:

$$m_{\text{req}} = \left\lceil \frac{\sigma_f^2}{\sigma_y^2} \cdot \frac{R_t^2}{R^2_{\text{clean}} - R_t^2} \right\rceil. \tag{6}$$

The two expressions have the same form—both invert "explained variance over total variance"—and differ only in which variance appears in the denominator and in the fact that the clean baseline of real data is not 1:

Of the four blinded targets, **two were met and two fell marginally short** (all shortfalls within ± 0.008). This is the same phenomenon as the negative deviation above: Eq. (4) accounts only for the structural attenuation caused by reference noise and *not* for the additional loss incurred when the model is retrained on noisy labels. We therefore **restate the status of the rule accurately as "a lower bound on the required number of replicates"** and give an *advisory* margin rule: discount R^2_{clean} by 1% before inverting Eq. (6). Under that correction (the per-target inversion and lookup are in the companion workbook), two things must be distinguished. First, **two of the four blinded targets have their m_{req} unchanged** (DM 0.75 stays at $m = 6$, SSC 0.75 stays at $m = 4$); both already met their targets before the correction (0.7536, 0.7578) and therefore *do not test the margin rule at all*. Second, only two targets are actually raised by the rule—exactly the two that previously fell short. **SSC 0.80 rises from $m = 7$ to $m = 8$, where the measured value is 0.8062, turning a shortfall into a pass; this is the one occasion on which the margin rule is directly verified by measurement.** DM 0.78 rises from $m = 9$ to $m = 11$, a level that **lies outside the grid (1–$10, 14, 20, \infty$) and therefore has no measured value**. We do *not* interpolate and report only the neighbouring levels—$m = 10$ measures 0.7794 (still 0.0006 short) and $m = 14$ measures 0.7911 (meets the target). **The rule is thus directly verified in exactly one case and merely bracketed in the other**, with the remaining two targets not participating in the test. The 1% figure is accordingly an **empirical margin induced from these four points, not a universal constant**; its evidential basis is one direct verification plus one bracketing, and it must be recalibrated on other data.

This section **does not claim** that the demonstration is equivalent to validation on the apple data. What it establishes is narrower: when the spectral signal is strong enough, reference noise limits performance in the predicted way. It cannot conversely explain the weak performance on the apple data. For that, §4.3 offers limited acquisition configuration as the hypothesis most consistent with the available diagnostics—**a hypothesis, not a tested cause**: we ran no experiment that varies the acquisition configuration, and we did not individually exclude alternatives such as model class. Scaling the apple variance ratio onto kiwifruit merely gives the demonstration a noise magnitude with an empirical basis; it is not an assertion that the kiwifruit reference determinations carry variability of that size.

The relation between this section and the attenuation-ratio test of §3.2 should be stated explicitly: **the two bracket the same proposition from opposite sides**, rather than one being strong and the other weak. Section 3.2 uses *real* replicate reference determinations (five destructive determinations per apple) but a weak model ($R^2 \approx 0.06$–0.12); its limitation is power (an interval of ± 0.03), not validity—the attenuation

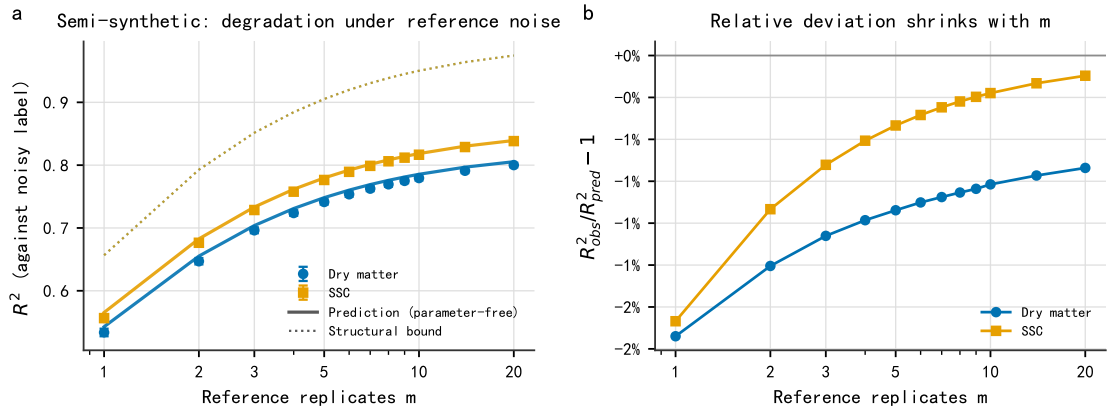

Figure 5: Semi-synthetic reference-noise degradation experiment. (a) Measured R^2 (points, error bars are seed-clustered bootstrap 95% CIs) against the prediction of Eq. (5) (solid) and the structural ceiling (dashed) as a function of the number of reference replicates m. (b) Shrinkage of the relative deviation $R^2_{\text{obs}}/R^2_{\text{pred}} - 1$ with m: uniformly negative and monotone, consistent with the mechanism that the extra retraining loss falls as the injected noise falls.

prediction does not require the model to be good, only that it perceive whole-fruit information alone, and the weak apple spectra make that premise more credible, not less. This section is the converse: the model is strong (0.827/0.861) but the replicates are *synthetic*, and the injected noise is i.i.d. by construction, so the most fragile premise of Eq. (2)—face-level deviations mutually uncorrelated and uncorrelated with the spectral signal—is supplied by us rather than tested. The systematic weaknesses are therefore **orthogonal**: real replicates with a weak model, versus a strong model with synthetic replicates. Our argument requires both to hold and rests on neither alone. The failure mode most likely to occur in practice—the tissue sampled destructively being the same tissue the effective optical sampling depth probes, making "reference noise" correlated with what the model can perceive—is excluded by neither side, and needs a dataset carrying both real replicate reference determinations and a working spectral model (§4.3(4)).

4. Discussion

4.1. How the existing literature should be read

It must be stated clearly that **this paper does not claim that the high R^2 values reported in the literature are wrong**. The split-leakage magnitude we measure is 1.41–1.46×, which is not enough to explain the gap to typical published values; and our own apple data already perform poorly on the spectral side under the protocols evaluated (§4.3), so we are in no position to judge anyone else's model performance. Our claim is much weaker and much more robust: when the reference value is a single local determination, the ceiling of Eq. (2) holds for **any predictor that can perceive only whole-fruit information**—provided the mean-independence condition of §2.2 holds, which is what makes the statement model-free rather than linear-only—and in that sense is a structural constraint that does not depend on the particular model form (regressor, preprocessing, hyperparameters). **Its condition must be stated precisely**: if spectra and reference determinations are paired on the same face and local chemical differences do change the local spectrum, then a model can in principle predict part of the biological content of e_{ij}, and Eq. (2) then ceases to be a ceiling for every model and becomes a conditional ceiling for whole-fruit-level predictors only. Our apple data cannot distinguish the two cases, since they contain no technical replicates on the same face (§4.3(2)); we therefore assert the ceiling only under that condition. Whether a given study is bound by the constraint depends on how far its performance sits from the ceiling—something the attenuation diagnostic given here can check directly on that study's own data.

It is worth stressing that the ceiling is not binding for every study. Our apple data measure $R^2 = 0.118$ **on the whole-fruit mean target**, so the ceiling to read it against is the five-face one, 0.820, not the single-face 0.477; it is far below either, which shows that reference-value variability is not the binding constraint on that dataset's current performance. What does bind it we do not establish: "limited acquisition configuration" is the hypothesis most consistent with the available diagnostics, but **no experiment in this paper varies the acquisition configuration**, and alternatives such as model class or parameter space were not individually excluded—so this is a candidate explanation, not a demonstrated cause. The ceiling becomes a real constraint only for studies that approach it.

4.2. Practical implications of the two components

On the reference side, the choice follows from the research target. If that target is whole-fruit quality, the reference value should be a multi-point mean, and the required number of replicates can be fixed with the design table of §3.6. If the target is local quality, spectra and reference determinations must be paired at the same location, and the result must not be evaluated against a whole-fruit label.

On the measurement side, an RMSEP reported on a single instrument omits the part of the error that appears only on cross-instrument deployment. At the two operating points compared here, the better the model, the larger that part's share of the total error (§3.4). Performance reports should therefore give both an "instrument-invariant part" and a "changes-with-instrument part". Calibration-transfer methods target the correctable device main effect [21, 22, 23, 3, 24], whereas the fruit×device residual component cannot be separated from scan noise in the absence of technical replicates on the same device and is hard to remove by post-processing. Similar transfer strategies have been extended to other spectroscopic applications [44], but they too address correctable systematic differences.

4.3. Limitations

(1) The spectral side of the apple data performs poorly under the protocols evaluated here. Under fruit-grouped holdout, the R^2 for whole-fruit mean SSC is only about 0.11. That figure comes from **two** independently written evaluation implementations: the main implementation gives 0.118 (95% CI [0.109, 0.127], component count searched over 1–20, 20 repeats per seed) and the other, with a different component grid and repeat count (2–24 in steps of 2, 15 repeats per seed), gives 0.109 (95% CI [0.095, 0.121]). The two intervals overlap strongly, which supports only "the conclusion is insensitive to implementation details and to the random order of splitting"; it **does not constitute replication on independent data**. Either way, both are far below that dataset's five-face-mean ceiling of 0.820—the one that matches this target—and the conclusion is unaffected. Three diagnostics *failed to support* the obvious explanations.

1. *Not the splitting scheme.* A row-wise random split gives only 0.169.
2. *Not saturation or band selection.* Mean saturation over the third overtone band of sugar (900–980 nm) is only 1.2%. Under that second implementation the full spectrum gives 0.109; restricting to non-saturated bands is worse, at 0.077; and 900–1001 nm alone is lower still, at 0.015.
3. *No sign of a general modelling failure.* The same code and protocol give $R^2 = 0.827$ for DM and $R^2 = 0.861$ for SSC on the kiwifruit data. That protocol is structurally identical to the apple analysis—fruit-grouped repeated holdout, 4-fold inner grouped cross-validation for component selection—but uses 6 repeats per seed against 20 for the apple main table and 15 for the band subsets. It is also a different evaluation regime from the $R^2 = 0.819$ obtained under 5-fold grouped cross-validation in §3.4; **the two must not be mixed**.

The single band most correlated with SSC is at 678 nm ($|r| = 0.154$), in the chlorophyll absorption and red-edge region, which is a colour–maturity proxy rather than direct sugar absorption. The reasonable attribution is that reflectance geometry on intact fruit has an effective sampling depth of only millimetres and mainly probes the skin and subcutaneous tissue [45, 46, 47, 48], whereas the refractometer measures juice from destructively sampled flesh; and that 391–1001 nm covers only the very weak third overtone, with the strong sugar absorption bands lying outside the range. This paper therefore **makes no claim whatsoever about a better spectral estimator**.

(2) The source of the within-fruit variability cannot be fully decomposed. This dataset has no technical replicates on the same sampling face, so genuine spatial chemical heterogeneity cannot be separated from the variability of destructive sampling and sample preparation. We use the term **within-fruit reference variability** throughout. Nothing in the practical conclusions is lost by this, and the reason should be argued rather than asserted. Write the two sources as genuine spatial chemical heterogeneity and sampling-and-preparation error. **So long as the face-level departures are approximately independent across faces, both decay as $1/m$ alike**: Eqs. (2) and (4) are unchanged word for word, and only the accompanying recommendation changes (besides raising the number of replicates, one should also standardise the sampling procedure). That independence is precisely what §3.2 tests: the permutation test of compound symmetry cannot be rejected ($p = 0.261$). Conversely, a systematic procedural bias acting at the *whole-fruit* level is by definition absorbed into σ_a^2, which can only **raise** the ICC and make the ceiling look more permissive — it cannot make our conclusions optimistic. Whether σ_f^2 is biological heterogeneity or procedural error therefore affects where the improvement effort should go, not whether the design formula holds. Decomposing the source would require replicate determinations on the same face, i.e. new acquisition. To let the reader judge the boundary of this reading, we state what the acquisition record **does** and **does not** contain. It *does* define the sampling faces (top, bottom, and three sides taken every 120° counterclockwise around the equatorial band) and the procedure that the refractometer is read after destructive sampling. It *does not* record the mass of tissue taken per face, whether each face was juiced separately, or whether the readings were temperature-compensated. Those three are precisely what would be needed to separate genuine spatial chemical heterogeneity from sampling-and-preparation error, so we make no claim about their relative shares.

(3) The instrument component is a pooled term. The kiwifruit data contain no technical replicates within a device, so "instrument×sample interaction" and "scan/repositioning noise" are inseparable; we report only the pooled term and do not claim to have estimated pure instrument repeatability. A second coverage limit applies on this side: the 1068–1137 nm columns are missing on *all five* devices, so the instrument budget is estimated only over 402–1065 nm and says nothing about the region beyond it.

(4) The two components come from different species; this is a division of labour, not mutual corroboration. The reference side comes from apples and the measurement side from kiwifruit; they cannot validate each other and each stands on its own. The reason for placing them side by side is that no public dataset currently supplies both reference replicates and cross-instrument spectral replicates.

(5) A single collection. The apple data come from one 2025 collection across three origins (the acquisition record states only the year and the origins; whether it constitutes a single acquisition batch is **not recorded**) and the ICC varies with origin (0.335–0.615), so the ceiling must be reported per collection and cannot be used as a species constant.

(6) Two preprocessing masks are computed before splitting. A paper about validation protocol should state where its own protocol falls short of the standard it argues for. Two feature masks in our pipeline are built from the *whole* dataset and only then fed to the grouped holdouts: the saturation mask behind the band-subset diagnostic (`A6formal_v18.py`, retaining wavelengths whose combined under/over-range rate is $\leq 1\%$), and the missing-channel mask for the kiwifruit model (`A6formal_v18.py`, dropping wavelength columns absent for some devices). Both are **target-blind**—they read only the spectra, never the reference values—and dropping channels fold-by-fold would give the folds different feature spaces and make them incomparable; but under a strict no-clean-separation criterion [25] they are still decisions taken with held-out spectra in view. We report them rather than re-derive them, and note that the affected quantities are the band-subset diagnostic and the kiwifruit R^2 of 0.827/0.861 used as the clean baseline in §3.7. For the kiwifruit mask the question can be settled rather than merely bounded. After the rows lacking a reference value are dropped, **the spectral block contains no missing value at all**: the mask removes **0 of 222** wavelength columns. A mask recomputed inside a training split can only ever keep a superset of the whole-data mask—fewer rows mean fewer opportunities for a column to be missing—so once the whole-data mask keeps every column, any fold-local mask is **necessarily identical, for every conceivable split**. We also checked this directly on the splits actually scored: recomputing the mask inside each of the $5 \times 6 = 30$ outer training splits of the reported kiwifruit run reproduces it 30/30 times. Its position relative to the split therefore cannot affect the evaluated feature space, the predictions or the scores. Separately, one exploratory script selects a preprocessing method by held-out R^2; **no number from that selection enters this paper**, and we flag it because the released code contains it.

5. Conclusions

The RMSEP reported in non-destructive fruit quality assessment contains two variance components that are not the model's doing. This paper gives a workflow for calibrating each of them separately and converts the result into a design rule usable in advance, $m \geq (\sigma_f^2/\sigma_a^2) \cdot R^2_{\max}/(1 - R^2_{\max})$. Calibrated on our data, the two components come out as follows. On the reference side, within-individual variability puts the attainable R^2 ceiling for apple SSC at 0.477 under a single-face reference value; across the three origins in this collection that ceiling varies between 0.335 and 0.615. On the measurement side, the pooled device×sample and scan/repositioning component corresponds to about 0.49 %DM of RMSEP in kiwifruit dry-matter prediction. That absolute quantity is approximately invariant between the two model operating points compared here, so under the strong model it accounts for 24.6% of the mean squared prediction error.

The variance-component model underpinning these conclusions yields four groups totalling six parameter-free point predictions whose measured values are all numerically close to theory (the label-only baseline, for instance, has theoretical value 0.3462 against a measured 0.3456), although no formal pass threshold was set for any of them except exchangeability across faces. On that basis we deliver two tools: a design table for the number of reference replicates, and an attenuation-ratio diagnostic that any laboratory can run on its own data to check whether it is bound by the limit. Both presuppose a leakage-free validation protocol, so we also specify a two-level grouped one *together with its audit procedure.*

The protocol itself—group by individual, then by acquisition day—is the deliverable; whether each level is actually necessary must be audited on the data at hand. In our apple data the individual level was confirmed necessary. An apparently deeper acquisition-day-level leakage in the time-series data was, by contrast, withdrawn after a control test: the controls do not support a day-level leakage mechanism that stands on its own, and the drop is mainly attributable to a target derived from the grouping variable. One storage-duration combination still meets the tightened criterion, so a small genuine same-day shared-condition effect cannot be excluded. We report that analysis in the Supplementary Material as an instance of that class of trap.

The spectral side of the apple data performs poorly under the PLS protocols evaluated here, so the ceiling was never approached on that dataset. We therefore ran a pre-registered semi-synthetic experiment (§3.7) on the kiwifruit data, whose spectral signal is strong enough: reference noise scaled by the variance ratio measured on apples was injected into the labels, and the model retrained. The degradation trajectory agrees with the parameter-free prediction at all 13 replicate levels (median signed relative deviations of -0.9% and -0.3%) and never crosses the structural ceiling. That experiment also exposed a real boundary of the design rule—it accounts only for structural attenuation and not for the additional loss of retraining on noisy reference values—so it should be used as a *lower bound*. The 1% empirical margin proposed on that basis changes the requirement for only two of the four blinded targets: one (SSC 0.80, m from 7 to 8) is directly verified by measurement, while the other (DM 0.78, m from 9 to 11) is only bracketed by adjacent levels because the level it requires lies outside the experimental grid. The margin is therefore an empirical value, not a universal constant.

Data and code availability

The apple data are unpublished primary research material held by Xinjiang Agricultural University; its data-sharing rules reserve the release decision to it, so the data are not deposited in a public repository and are available from the corresponding author on reasonable request. The kiwifruit data are public and were used exactly as deposited: Kiwifruit NIR, figshare, v1, CC BY 4.0, https://doi.org/10.6084/m9.figshare.21747983.v1 [30], first described by Wohlers et al. [31].

All analysis scripts, intermediate result workbooks, per-run records, figures and the pre-registered statistical protocol are released at https://github.com/2004lryan/020nir-reference-error-budget, together with a claim-by-claim map, so every statistic reported here is checkable without access to the raw data.

CRediT authorship contribution statement

Panlin Li: Methodology, Software, Formal analysis, Visualization, Writing – original draft. **Yuchang Li**: Investigation, Data curation, Validation. **Yutong Feng**: Investigation, Data curation. **Longjie Li**: Resources, Investigation. **Ya Liu**: Resources, Investigation. **Hua Huang**: Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude in order to assist with language polishing and code review. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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